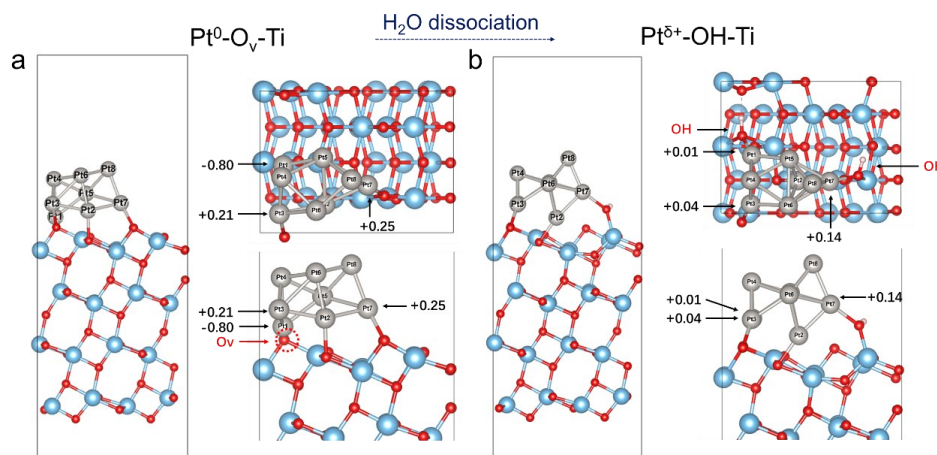


MAIN ARTICLE - REMOVED ORIGINAL CONTENT

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active in activating CO molecules. It was noteworthy that in bimolecular redox reactions, instantaneous dynamic changes in the oxidation state of active sites at the interface were ubiquitous. DFT calculations and Bader charge analysis revealed the charge on the Pt<sub>1</sub> atom in the Pt<sup>0</sup>-O<sub>v</sub>-Ti structure changed from  $-0.8 |e|$  to  $+0.01 |e|$  during the water dissociation process, indicating the formation of the Pt<sup>δ+</sup>-OH-Ti structure (Supplementary Fig. S25 and Table S2). This observation aligned with the dynamic evolution of the oxidation state of Pt species at the interface during the reaction.

SUPPLEMENTARY INFORMATION - REMOVED ORIGINAL



**Supplementary Fig. S25 | Dynamic changes in the oxidation state of Pt species at the interface.** The change of Bader charge for various Pt atoms in the Pt<sub>cluster</sub>/TiO<sub>2</sub> catalysts (a) with the oxygen vacancy and (b) after the H<sub>2</sub>O dissociation.

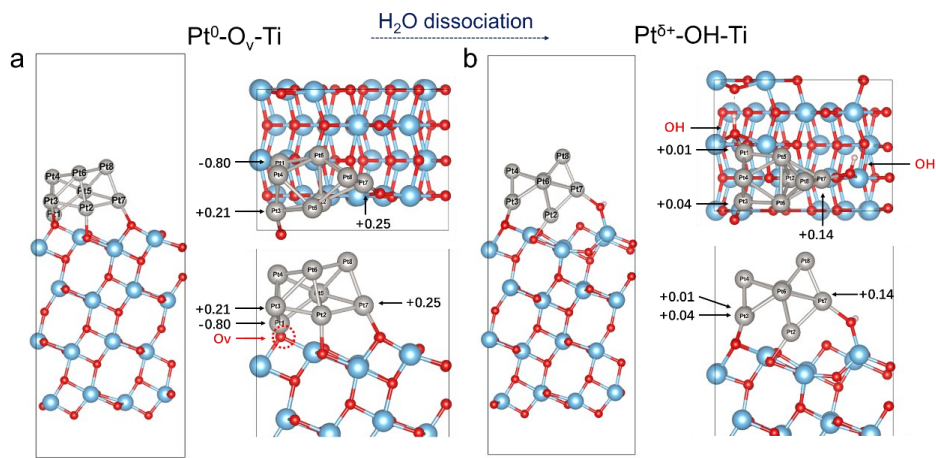
It was noteworthy that in bimolecular redox reactions, instantaneous dynamic changes in the oxidation state of active sites at the interface were ubiquitous. To reveal the dynamic changes in the state of Pt sites during the reaction, DFT calculations and Bader charge analysis were performed. As shown in Supplementary Fig. S25a, we constructed an eight-atom Pt cluster on the TiO<sub>2</sub>(101) surface and removed the oxygen atom at the interface bonded to the Pt<sub>1</sub> atom to create an oxygen vacancy (Vo). Analysis of the Bader charge revealed a value of -0.8 |e| for the Pt<sub>1</sub> atom, indicating that the Pt<sub>1</sub> atom was most likely in the metallic state (Supplementary Table S2). Therefore, the active structure identified in our experiments, denoted as Pt<sup>0</sup>-O<sub>v</sub>-Ti, has been successfully modeled. During the water dissociation process, the oxygen vacancy was filled by the -OH group. The Bader charge of Pt<sub>1</sub> changed to +0.01 |e|, confirming the formation of the Pt<sup>δ+</sup>-OH-Ti structure (Supplementary Fig. S25b). Notably, the other H\* atom bonded from H<sub>2</sub>O dissociation with a surface oxygen atom to form a new -OH group. Concurrently, the Bader charge of the adjacent Pt<sub>7</sub> atom decreased from +0.25 |e| to +0.14 |e|. This observation clearly demonstrated the dynamic evolution of the oxidation state of Pt species at the interface during the reaction.

**Supplementary Table 2.** The values of Bader charge for various Pt atoms in the Pt<sub>cluster</sub>/TiO<sub>2</sub> catalysts with the oxygen vacancy and after the H<sub>2</sub>O dissociation.

| Pt atoms   | Bader Charge        |                                     |
|------------|---------------------|-------------------------------------|
|            | With oxygen vacancy | After H <sub>2</sub> O dissociation |
| <b>Pt1</b> | <b>-0.80</b>        | <b>0.01</b>                         |
| Pt2        | 0.09                | -0.27                               |
| <b>Pt3</b> | <b>0.21</b>         | <b>0.04</b>                         |
| Pt4        | -0.10               | -0.10                               |
| Pt5        | -0.04               | 0.06                                |
| Pt6        | 0.02                | 0.06                                |
| <b>Pt7</b> | <b>0.25</b>         | <b>0.14</b>                         |
| Pt8        | -0.10               | -0.11                               |

RESPONSE LETTER - REMOVED ORIGINAL CONTENT

**Response:** We thank the reviewer for the valuable comments. In bimolecular redox reactions, the instantaneous dynamic change of the active site oxidation state was ubiquitous (Li, Y. et al. *Nat. Common.* **2021**, 12, 914). To reveal the dynamic changes in the state of Pt sites during the reaction, DFT calculations and Bader charge analysis were performed (Figure R2-12 and Table R2-2). As shown in Figure R2-12a, we successfully constructed an eight-atom Pt cluster on the  $\text{TiO}_2(101)$  surface and removed the oxygen atom at the interface bonded to the  $\text{Pt}_1$  atom to create an oxygen vacancy ( $\text{V}_\text{o}$ ). Analysis of the Bader charge revealed a value of  $-0.8 |e|$  for the  $\text{Pt}_1$  atom, indicating that the  $\text{Pt}_1$  atom was most likely in the metallic state. Therefore, the active structure identified in our experiments, denoted as  $\text{Pt}^0\text{-O}_\text{v}\text{-Ti}$ , has been successfully modeled. During the water dissociation process, the oxygen vacancy was filled by the  $\text{-OH}$  group, and the Bader charge of  $\text{Pt}_1$  changes to  $+0.01 |e|$ , confirming the formation of the  $\text{Pt}^{\delta+}\text{-OH-Ti}$  structure (Figure R2-12b). Notably, the other  $\text{H}^*$  atom bonds from  $\text{H}_2\text{O}$  dissociation with a surface oxygen atom to form a new  $\text{-OH}$  group. Concurrently, the Bader charge of the adjacent  $\text{Pt}_7$  atom decreased from  $+0.25 |e|$  to  $+0.14 |e|$ . This observation clearly demonstrated the dynamic evolution of the oxidation state of Pt species at the interface during the reaction. **Figure R2-12 has been shown as Figure S25 in the supplementary information. Table R2-2 has been shown as Table S2 in the supplementary information. The corresponding description has been added on Page 11, lines 10–16 (highlight in yellow).**



**Figure R2-12.** The change of Bader charge for various Pt atoms in the Pt<sub>cluster</sub>/TiO<sub>2</sub> catalysts (a) with the oxygen vacancy and (b) after the H<sub>2</sub>O dissociation.

**Table R2-2.** The values of Bader charge for various Pt atoms in the Pt<sub>cluster</sub>/TiO<sub>2</sub> catalysts with the oxygen vacancy and after the H<sub>2</sub>O dissociation.

| Pt atoms | Bader Charge        |                                     |
|----------|---------------------|-------------------------------------|
|          | With oxygen vacancy | After H <sub>2</sub> O dissociation |
| Pt1      | -0.80               | 0.01                                |
| Pt2      | 0.09                | -0.27                               |
| Pt3      | 0.21                | 0.04                                |
| Pt4      | -0.10               | -0.10                               |
| Pt5      | -0.04               | 0.06                                |
| Pt6      | 0.02                | 0.06                                |
| Pt7      | 0.25                | 0.14                                |
| Pt8      | -0.10               | -0.11                               |